

## T3 group factors: UV

Active theory: SM +  $S_1 + S_2 + F$ . The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table,  $\mathcal{S}$  denotes the RGE structure multiplying the displayed group factor in  $16\pi^2\beta$ .

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q}Y_u\tilde{H}u + \text{h.c.}$                              |
| $Y_d$                     | $-\bar{Q}Y_d Hd + \text{h.c.}$                                     |
| $Y_e$                     | $-\bar{L}Y_e He + \text{h.c.}$                                     |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F}M_F F$                                                    |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$                             |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$               |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$               |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$              |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$              |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$      |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$      |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$            |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$    |
| $\lambda_{12}^{(\times)}$ | crossed independent $S_1$ – $S_2$ contraction                      |
| $\lambda_{S_1}^{(A)}$     | additional independent $S_1$ self-contraction                      |
| $\lambda_{S_2}^{(A)}$     | additional independent $S_2$ self-contraction                      |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2-1)/4$                                                        |
| $T(d)$                    | $d(d^2-1)/12$                                                      |
| $C_{LLS_1S_2}^{ij:AB}$    | $L_i L_j S_1^A S_2^B$                                              |
| $C_{ijab}$                | $L_i L_j \phi_a \phi_b$                                            |
| $C_5^{ij}$                | $(L_i^T C L_j) H H$                                                |
| $R_W$                     | $16\pi^2\beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$            |

## Field representations

| run    | $S_1$     | $S_2$    | $F$                 | $\alpha$ |
|--------|-----------|----------|---------------------|----------|
| T3 – A | $(1, -1)$ | $(3, 0)$ | $(2, -\frac{1}{2})$ | $-2$     |

|        |                     |                    |                     |    |
|--------|---------------------|--------------------|---------------------|----|
| T3 – A | $(1, -\frac{1}{2})$ | $(3, \frac{1}{2})$ | $(2, 0)$            | -1 |
| T3 – A | $(1, 0)$            | $(3, 1)$           | $(2, \frac{1}{2})$  | 0  |
| T3 – A | $(1, \frac{1}{2})$  | $(3, \frac{3}{2})$ | $(2, 1)$            | 1  |
| T3 – A | $(1, 1)$            | $(3, 2)$           | $(2, \frac{3}{2})$  | 2  |
| T3 – B | $(2, -1)$           | $(2, 0)$           | $(1, -\frac{1}{2})$ | -2 |
| T3 – B | $(2, -\frac{1}{2})$ | $(2, \frac{1}{2})$ | $(1, 0)$            | -1 |
| T3 – B | $(2, 0)$            | $(2, 1)$           | $(1, \frac{1}{2})$  | 0  |
| T3 – B | $(2, \frac{1}{2})$  | $(2, \frac{3}{2})$ | $(1, 1)$            | 1  |
| T3 – B | $(2, 1)$            | $(2, 2)$           | $(1, \frac{3}{2})$  | 2  |
| T3 – C | $(2, -1)$           | $(2, 0)$           | $(3, -\frac{1}{2})$ | -2 |
| T3 – C | $(2, -\frac{1}{2})$ | $(2, \frac{1}{2})$ | $(3, 0)$            | -1 |
| T3 – C | $(2, 0)$            | $(2, 1)$           | $(3, \frac{1}{2})$  | 0  |
| T3 – C | $(2, \frac{1}{2})$  | $(2, \frac{3}{2})$ | $(3, 1)$            | 1  |
| T3 – C | $(2, 1)$            | $(2, 2)$           | $(3, \frac{3}{2})$  | 2  |
| T3 – D | $(3, -1)$           | $(1, 0)$           | $(2, -\frac{1}{2})$ | -2 |
| T3 – D | $(3, -\frac{1}{2})$ | $(1, \frac{1}{2})$ | $(2, 0)$            | -1 |
| T3 – D | $(3, 0)$            | $(1, 1)$           | $(2, \frac{1}{2})$  | 0  |
| T3 – D | $(3, \frac{1}{2})$  | $(1, \frac{3}{2})$ | $(2, 1)$            | 1  |
| T3 – D | $(3, 1)$            | $(1, 2)$           | $(2, \frac{3}{2})$  | 2  |
| T3 – E | $(3, -1)$           | $(3, 0)$           | $(2, -\frac{1}{2})$ | -2 |
| T3 – E | $(3, -\frac{1}{2})$ | $(3, \frac{1}{2})$ | $(2, 0)$            | -1 |
| T3 – E | $(3, 0)$            | $(3, 1)$           | $(2, \frac{1}{2})$  | 0  |
| T3 – E | $(3, \frac{1}{2})$  | $(3, \frac{3}{2})$ | $(2, 1)$            | 1  |
| T3 – E | $(3, 1)$            | $(3, 2)$           | $(2, \frac{3}{2})$  | 2  |

## Gauge and Yukawa beta functions

$\beta_{g_1}^{(1)}$

| $\mathcal{S}$ | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^3$       | $\frac{55}{6}$           | $\frac{43}{6}$           | $\frac{59}{6}$          | $\frac{103}{6}$         | $\frac{175}{6}$         |

| $\mathcal{S}$ | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^3$       | $\frac{17}{2}$           | $\frac{43}{6}$           | $\frac{17}{2}$          | $\frac{25}{2}$          | $\frac{115}{6}$         |

| $\mathcal{S}$ | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^3$       | $\frac{21}{2}$           | $\frac{43}{6}$           | $\frac{21}{2}$          | $\frac{41}{2}$          | $\frac{223}{6}$         |

| $\mathcal{S}$ | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^3$       | $\frac{59}{6}$           | $\frac{43}{6}$           | $\frac{55}{6}$          | $\frac{95}{6}$          | $\frac{163}{6}$         |

| $\mathcal{S}$ | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^3$       | $\frac{59}{6}$           | $\frac{22}{3}$           | $\frac{59}{6}$          | $\frac{52}{3}$          | $\frac{179}{6}$         |

$\beta_{g_2}^{(1)}$

|                               |                          |                          |                         |                         |                         |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\mathcal{S}$                 | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
| $g_2^3$                       | $-\frac{1}{2}$           | $-\frac{3}{2}$           | $-\frac{1}{2}$          | $-\frac{1}{2}$          | $-\frac{1}{2}$          |
| $\beta_{g_2}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
| $g_2^3$                       | $-\frac{17}{6}$          | $-\frac{17}{6}$          | $-\frac{17}{6}$         | $-\frac{17}{6}$         | $-\frac{17}{6}$         |
| $\beta_{g_2}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
| $g_2^3$                       | $\frac{31}{6}$           | $\frac{7}{6}$            | $\frac{31}{6}$          | $\frac{31}{6}$          | $\frac{31}{6}$          |
| $\beta_{g_2}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
| $g_2^3$                       | $-\frac{1}{2}$           | $-\frac{3}{2}$           | $-\frac{1}{2}$          | $-\frac{1}{2}$          | $-\frac{1}{2}$          |
| $\beta_{g_2}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
| $g_2^3$                       | $\frac{1}{6}$            | $-\frac{5}{6}$           | $\frac{1}{6}$           | $\frac{1}{6}$           | $\frac{1}{6}$           |
| $\beta_{g_2}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
| $g_3^3$                       | $-7$                     | $-7$                     | $-7$                    | $-7$                    | $-7$                    |
| $\beta_{g_3}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
| $g_3^3$                       | $-7$                     | $-7$                     | $-7$                    | $-7$                    | $-7$                    |
| $\beta_{g_3}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
| $g_3^3$                       | $-7$                     | $-7$                     | $-7$                    | $-7$                    | $-7$                    |
| $\beta_{g_3}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
| $g_3^3$                       | $-7$                     | $-7$                     | $-7$                    | $-7$                    | $-7$                    |
| $\beta_{g_3}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
| $g_3^3$                       | $-7$                     | $-7$                     | $-7$                    | $-7$                    | $-7$                    |
| $\beta_{g_3}^{(1)}$           |                          |                          |                         |                         |                         |
| $\mathcal{S}$                 | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | $-8$                     | $-8$                     | $-8$                    | $-8$                    | $-8$                    |

|                   |                  |                  |                  |                  |                  |
|-------------------|------------------|------------------|------------------|------------------|------------------|
| $g_1^2(Y_u)_{ij}$ | $-\frac{17}{12}$ | $-\frac{17}{12}$ | $-\frac{17}{12}$ | $-\frac{17}{12}$ | $-\frac{17}{12}$ |
| $T(Y_u)_{ij}$     | 1                | 1                | 1                | 1                | 1                |

| $S$                           | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | -8                       | -8                       | -8                      | -8                      | -8                      |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$         | $-\frac{17}{12}$         | $-\frac{17}{12}$        | $-\frac{17}{12}$        | $-\frac{17}{12}$        |
| $T(Y_u)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $S$                           | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | -8                       | -8                       | -8                      | -8                      | -8                      |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$         | $-\frac{17}{12}$         | $-\frac{17}{12}$        | $-\frac{17}{12}$        | $-\frac{17}{12}$        |
| $T(Y_u)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $S$                           | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | -8                       | -8                       | -8                      | -8                      | -8                      |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$         | $-\frac{17}{12}$         | $-\frac{17}{12}$        | $-\frac{17}{12}$        | $-\frac{17}{12}$        |
| $T(Y_u)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $S$                           | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | -8                       | -8                       | -8                      | -8                      | -8                      |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$         | $-\frac{17}{12}$         | $-\frac{17}{12}$        | $-\frac{17}{12}$        | $-\frac{17}{12}$        |
| $T(Y_u)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

$\beta_{Y_d}^{(1)}$

| $S$                           | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_d)_{ij}$             | -8                       | -8                       | -8                      | -8                      | -8                      |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$          | $-\frac{5}{12}$          | $-\frac{5}{12}$         | $-\frac{5}{12}$         | $-\frac{5}{12}$         |
| $T(Y_d)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $S$ | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-----|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
|-----|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|

|                               |                 |                 |                 |                 |                 |
|-------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$  | $-\frac{3}{2}$  | $-\frac{3}{2}$  | $-\frac{3}{2}$  | $-\frac{3}{2}$  |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$  | $-\frac{9}{4}$  | $-\frac{9}{4}$  | $-\frac{9}{4}$  | $-\frac{9}{4}$  |
| $g_3^2(Y_d)_{ij}$             | $-8$            | $-8$            | $-8$            | $-8$            | $-8$            |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$ | $-\frac{5}{12}$ | $-\frac{5}{12}$ | $-\frac{5}{12}$ | $-\frac{5}{12}$ |
| $T(Y_d)_{ij}$                 | $1$             | $1$             | $1$             | $1$             | $1$             |

| $\mathcal{S}$                 | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_d)_{ij}$             | $-8$                     | $-8$                     | $-8$                    | $-8$                    | $-8$                    |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$          | $-\frac{5}{12}$          | $-\frac{5}{12}$         | $-\frac{5}{12}$         | $-\frac{5}{12}$         |
| $T(Y_d)_{ij}$                 | $1$                      | $1$                      | $1$                     | $1$                     | $1$                     |

| $\mathcal{S}$                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_d)_{ij}$             | $-8$                     | $-8$                     | $-8$                    | $-8$                    | $-8$                    |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$          | $-\frac{5}{12}$          | $-\frac{5}{12}$         | $-\frac{5}{12}$         | $-\frac{5}{12}$         |
| $T(Y_d)_{ij}$                 | $1$                      | $1$                      | $1$                     | $1$                     | $1$                     |

| $\mathcal{S}$                 | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$           | $-\frac{3}{2}$           | $-\frac{3}{2}$          | $-\frac{3}{2}$          | $-\frac{3}{2}$          |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_3^2(Y_d)_{ij}$             | $-8$                     | $-8$                     | $-8$                    | $-8$                    | $-8$                    |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$          | $-\frac{5}{12}$          | $-\frac{5}{12}$         | $-\frac{5}{12}$         | $-\frac{5}{12}$         |
| $T(Y_d)_{ij}$                 | $1$                      | $1$                      | $1$                     | $1$                     | $1$                     |

$\beta_{Y_e}^{(1)}$

| $\mathcal{S}$                 | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          | $-\frac{15}{4}$          | $-\frac{15}{4}$         | $-\frac{15}{4}$         | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | $1$                      | $1$                      | $1$                     | $1$                     | $1$                     |

| $\mathcal{S}$                 | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          | $-\frac{15}{4}$          | $-\frac{15}{4}$         | $-\frac{15}{4}$         | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | $1$                      | $1$                      | $1$                     | $1$                     | $1$                     |

| $\mathcal{S}$                 | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          | $-\frac{15}{4}$          | $-\frac{15}{4}$         | $-\frac{15}{4}$         | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $\mathcal{S}$                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          | $-\frac{15}{4}$          | $-\frac{15}{4}$         | $-\frac{15}{4}$         | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

| $\mathcal{S}$                 | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          | $-\frac{15}{4}$          | $-\frac{15}{4}$         | $-\frac{15}{4}$         | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |

$\beta_{y_1}^{(1)}$

| $\mathcal{S}$                           | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(1)}(y_1)_{ij}$                 | 2                        | 2                        | 2                       | 2                       | 2                       |
| $(y_2(y_2)^\dagger y_1)_{ij}$           | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_1(y_1)^\dagger y_1)_{ij}$           | 1                        | 1                        | 1                       | 1                       | 1                       |
| $g_2^2(y_1)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_1)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_1(y_2)^\dagger y_2)_{ij}$           | 0                        | $\frac{3}{4}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

| $\mathcal{S}$                           | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(1)}(y_1)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_2(y_2)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_1(y_1)^\dagger y_1)_{ij}$           | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_1)_{ij}$                       | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(y_1)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_1(y_2)^\dagger y_2)_{ij}$           | 0                        | 1                        | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$ | 0                        | 1                        | 0                       | 0                       | 0                       |

| $\mathcal{S}$ | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|

|                                         |                 |                 |                 |                 |                 |
|-----------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| $T_\nu^{(1)}(y_1)_{ij}$                 | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   | $\frac{3}{2}$   |
| $(y_2(y_2)^\dagger y_1)_{ij}$           | $\frac{3}{4}$   | $\frac{3}{4}$   | $\frac{3}{4}$   | $\frac{3}{4}$   | $\frac{3}{4}$   |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$           | $\frac{1}{2}$   | $\frac{1}{2}$   | $\frac{1}{2}$   | $\frac{1}{2}$   | $\frac{1}{2}$   |
| $(y_1(y_1)^\dagger y_1)_{ij}$           | $\frac{5}{4}$   | $\frac{5}{4}$   | $\frac{5}{4}$   | $\frac{5}{4}$   | $\frac{5}{4}$   |
| $g_2^2(y_1)_{ij}$                       | $-\frac{33}{4}$ | $-\frac{33}{4}$ | $-\frac{33}{4}$ | $-\frac{33}{4}$ | $-\frac{33}{4}$ |
| $g_1^2(y_1)_{ij}$                       | $-\frac{3}{2}$  | $-\frac{3}{4}$  | $-\frac{3}{2}$  | $-\frac{15}{4}$ | $-\frac{15}{2}$ |
| $(y_1(y_2)^\dagger y_2)_{ij}$           | 0               | $\frac{1}{2}$   | 0               | 0               | 0               |
| $\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$ | 0               | $\frac{3}{2}$   | 0               | 0               | 0               |

| $S$                                     | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(1)}(y_1)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_2(y_2)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_1(y_1)^\dagger y_1)_{ij}$           | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_1)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_1)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_1(y_2)^\dagger y_2)_{ij}$           | 0                        | $\frac{1}{2}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

| $S$                                     | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(1)}(y_1)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_2(y_2)^\dagger y_1)_{ij}$           | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_1(y_1)^\dagger y_1)_{ij}$           | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_1)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_1)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_1(y_2)^\dagger y_2)_{ij}$           | 0                        | $\frac{3}{4}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$ | 0                        | 1                        | 0                       | 0                       | 0                       |

$\beta_{y_2}^{(1)}$

| $S$                                     | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_1(y_1)^\dagger y_2)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger y_2)_{ij}$           | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_2)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_2)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_2(y_1)^\dagger y_1)_{ij}$           | 0                        | $\frac{1}{2}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

| $S$                           | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$       | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_1(y_1)^\dagger y_2)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger y_2)_{ij}$ | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_2)_{ij}$             | $-\frac{9}{4}$           | $-\frac{9}{4}$           | $-\frac{9}{4}$          | $-\frac{9}{4}$          | $-\frac{9}{4}$          |
| $g_1^2(y_2)_{ij}$             | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_2(y_1)^\dagger y_1)_{ij}$ | 0                        | 1                        | 0                       | 0                       | 0                       |

|                                         |   |   |   |   |   |
|-----------------------------------------|---|---|---|---|---|
| $\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$ | 0 | 1 | 0 | 0 | 0 |
|-----------------------------------------|---|---|---|---|---|

| $\mathcal{S}$                           | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$                 | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $(y_1(y_1)^\dagger y_2)_{ij}$           | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger y_2)_{ij}$           | $\frac{5}{4}$            | $\frac{5}{4}$            | $\frac{5}{4}$           | $\frac{5}{4}$           | $\frac{5}{4}$           |
| $g_2^2(y_2)_{ij}$                       | $-\frac{33}{4}$          | $-\frac{33}{4}$          | $-\frac{33}{4}$         | $-\frac{33}{4}$         | $-\frac{33}{4}$         |
| $g_1^2(y_2)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_2(y_1)^\dagger y_1)_{ij}$           | 0                        | $\frac{1}{2}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$ | 0                        | $\frac{3}{2}$            | 0                       | 0                       | 0                       |

| $\mathcal{S}$                           | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$                 | 2                        | 2                        | 2                       | 2                       | 2                       |
| $(y_1(y_1)^\dagger y_2)_{ij}$           | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger y_2)_{ij}$           | 1                        | 1                        | 1                       | 1                       | 1                       |
| $g_2^2(y_2)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_2)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_2(y_1)^\dagger y_1)_{ij}$           | 0                        | $\frac{3}{4}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

| $\mathcal{S}$                           | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-----------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$                 | 1                        | 1                        | 1                       | 1                       | 1                       |
| $(y_1(y_1)^\dagger y_2)_{ij}$           | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$           | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $(y_2(y_2)^\dagger y_2)_{ij}$           | $\frac{3}{2}$            | $\frac{3}{2}$            | $\frac{3}{2}$           | $\frac{3}{2}$           | $\frac{3}{2}$           |
| $g_2^2(y_2)_{ij}$                       | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2(y_2)_{ij}$                       | $-\frac{3}{2}$           | $-\frac{3}{4}$           | $-\frac{3}{2}$          | $-\frac{15}{4}$         | $-\frac{15}{2}$         |
| $(y_2(y_1)^\dagger y_1)_{ij}$           | 0                        | $\frac{3}{4}$            | 0                       | 0                       | 0                       |
| $\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$ | 0                        | 1                        | 0                       | 0                       | 0                       |

$\beta_{M_F}$

Run block 1

| $\mathcal{S}$          | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1^T y_1^*) M_F$    | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $M_F(y_2^\dagger y_2)$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $g_2^2 M_F$            | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 M_F$            | $-\frac{3}{2}$           | 0                        | $-\frac{3}{2}$          | -6                      | $-\frac{27}{2}$         |

Run block 2

| $\mathcal{S}$          | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1^T y_1^*) M_F$    | 1                        | 1                        | 1                       | 1                       | 1                       |
| $M_F(y_2^\dagger y_2)$ | 1                        | 1                        | 1                       | 1                       | 1                       |
| $g_2^2 M_F$            | 0                        | 0                        | 0                       | 0                       | 0                       |
| $g_1^2 M_F$            | $-\frac{3}{2}$           | 0                        | $-\frac{3}{2}$          | -6                      | $-\frac{27}{2}$         |

Run block 3

| $\mathcal{S}$          | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1^T y_1^*) M_F$    | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $M_F(y_2^\dagger y_2)$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $g_2^2 M_F$            | $-12$                    | $-12$                    | $-12$                   | $-12$                   | $-12$                   |
| $g_1^2 M_F$            | $-\frac{3}{2}$           | $0$                      | $-\frac{3}{2}$          | $-6$                    | $-\frac{27}{2}$         |

Run block 4

| $\mathcal{S}$          | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1^T y_1^*) M_F$    | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $M_F(y_2^\dagger y_2)$ | $\frac{1}{2}$            | $\frac{1}{2}$            | $\frac{1}{2}$           | $\frac{1}{2}$           | $\frac{1}{2}$           |
| $g_2^2 M_F$            | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 M_F$            | $-\frac{3}{2}$           | $0$                      | $-\frac{3}{2}$          | $-6$                    | $-\frac{27}{2}$         |

Run block 5

| $\mathcal{S}$          | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $(y_1^T y_1^*) M_F$    | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $M_F(y_2^\dagger y_2)$ | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $g_2^2 M_F$            | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 M_F$            | $-\frac{3}{2}$           | $0$                      | $-\frac{3}{2}$          | $-6$                    | $-\frac{27}{2}$         |

$\beta_{m_1^2}$

Run block 1: T3-A–T3-A

| $\mathcal{S}$                   | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_1^2 T_\nu^{(1)}$             | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $M_F^\dagger M_F T_\nu^{(1)}$   | $-8$                     | $-8$                     | $-8$                    | $-8$                    | $-8$                    |
| $\lambda_{S_1}^{(1)} m_1^2$     | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $g_2^2 m_1^2$                   | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |
| $g_1^2 m_1^2$                   | $-6$                     | $-\frac{3}{2}$           | $0$                     | $-\frac{3}{2}$          | $-6$                    |
| $\lambda_{12}^{(1)} m_2^2$      | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $\lambda_{S_1}^{(A)} m_1^2$     | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |
| $\lambda_{12}^{(\times)} m_2^2$ | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |

Run block 2: T3-B–T3-B

| $\mathcal{S}$                   | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_1^2 T_\nu^{(1)}$             | $2$                      | $2$                      | $2$                     | $2$                     | $2$                     |
| $M_F^\dagger M_F T_\nu^{(1)}$   | $-4$                     | $-16$                    | $-4$                    | $-4$                    | $-4$                    |
| $\lambda_{S_1}^{(1)} m_1^2$     | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $g_2^2 m_1^2$                   | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 m_1^2$                   | $-6$                     | $-\frac{3}{2}$           | $0$                     | $-\frac{3}{2}$          | $-6$                    |
| $\lambda_{12}^{(1)} m_2^2$      | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $\lambda_{S_1}^{(A)} m_1^2$     | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |
| $\lambda_{12}^{(\times)} m_2^2$ | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |

Run block 3: T3-C–T3-C

| $\mathcal{S}$                   | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_1^2 T_\nu^{(1)}$             | 3                        | 3                        | 3                       | 3                       | 3                       |
| $M_F^\dagger M_F T_\nu^{(1)}$   | -6                       | -24                      | -6                      | -6                      | -6                      |
| $\lambda_{S_1}^{(1)} m_1^2$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^2 m_1^2$                   | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 m_1^2$                   | -6                       | $-\frac{3}{2}$           | 0                       | $-\frac{3}{2}$          | -6                      |
| $\lambda_{12}^{(1)} m_2^2$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{S_1}^{(A)} m_1^2$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} m_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 4: T3-D-T3-D

| $\mathcal{S}$                   | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_1^2 T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $M_F^\dagger M_F T_\nu^{(1)}$   | -4                       | -4                       | -4                      | -4                      | -4                      |
| $\lambda_{S_1}^{(1)} m_1^2$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $g_2^2 m_1^2$                   | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 m_1^2$                   | -6                       | $-\frac{3}{2}$           | 0                       | $-\frac{3}{2}$          | -6                      |
| $\lambda_{12}^{(1)} m_2^2$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{S_1}^{(A)} m_1^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(\times)} m_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$                   | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_1^2 T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $M_F^\dagger M_F T_\nu^{(1)}$   | -4                       | -4                       | -4                      | -4                      | -4                      |
| $\lambda_{S_1}^{(1)} m_1^2$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $g_2^2 m_1^2$                   | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 m_1^2$                   | -6                       | $-\frac{3}{2}$           | 0                       | $-\frac{3}{2}$          | -6                      |
| $\lambda_{12}^{(1)} m_2^2$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{S_1}^{(A)} m_1^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(\times)} m_2^2$ | 4                        | 4                        | 4                       | 4                       | 4                       |

$\beta_{m_2^2}$

Run block 1: T3-A-T3-A

| $\mathcal{S}$                   | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_2^2 T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $M_F^\dagger M_F T_\nu^{(2)}$   | -4                       | -4                       | -4                      | -4                      | -4                      |
| $\lambda_{S_2}^{(1)} m_2^2$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $g_2^2 m_2^2$                   | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 m_2^2$                   | 0                        | $-\frac{3}{2}$           | -6                      | $-\frac{27}{2}$         | -24                     |
| $\lambda_{12}^{(1)} m_1^2$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{S_2}^{(A)} m_2^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(\times)} m_1^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 2: T3-B-T3-B

| $\mathcal{S}$                   | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_2^2 T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $M_F^\dagger M_F T_\nu^{(2)}$   | -4                       | -16                      | -4                      | -4                      | -4                      |
| $\lambda_{S_2}^{(1)} m_2^2$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^2 m_2^2$                   | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 m_2^2$                   | 0                        | $-\frac{3}{2}$           | -6                      | $-\frac{27}{2}$         | -24                     |
| $\lambda_{12}^{(1)} m_1^2$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{S_2}^{(A)} m_2^2$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} m_1^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 3: T3-C-T3-C

| $\mathcal{S}$                   | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_2^2 T_\nu^{(2)}$             | 3                        | 3                        | 3                       | 3                       | 3                       |
| $M_F^\dagger M_F T_\nu^{(2)}$   | -6                       | -24                      | -6                      | -6                      | -6                      |
| $\lambda_{S_2}^{(1)} m_2^2$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^2 m_2^2$                   | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 m_2^2$                   | 0                        | $-\frac{3}{2}$           | -6                      | $-\frac{27}{2}$         | -24                     |
| $\lambda_{12}^{(1)} m_1^2$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{S_2}^{(A)} m_2^2$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} m_1^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 4: T3-D-T3-D

| $\mathcal{S}$                   | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_2^2 T_\nu^{(2)}$             | 4                        | 4                        | 4                       | 4                       | 4                       |
| $M_F^\dagger M_F T_\nu^{(2)}$   | -8                       | -8                       | -8                      | -8                      | -8                      |
| $\lambda_{S_2}^{(1)} m_2^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 m_2^2$                   | 0                        | 0                        | 0                       | 0                       | 0                       |
| $g_1^2 m_2^2$                   | 0                        | $-\frac{3}{2}$           | -6                      | $-\frac{27}{2}$         | -24                     |
| $\lambda_{12}^{(1)} m_1^2$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{S_2}^{(A)} m_2^2$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} m_1^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$                   | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $m_2^2 T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $M_F^\dagger M_F T_\nu^{(2)}$   | -4                       | -4                       | -4                      | -4                      | -4                      |
| $\lambda_{S_2}^{(1)} m_2^2$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $g_2^2 m_2^2$                   | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 m_2^2$                   | 0                        | $-\frac{3}{2}$           | -6                      | $-\frac{27}{2}$         | -24                     |
| $\lambda_{12}^{(1)} m_1^2$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{S_2}^{(A)} m_2^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(\times)} m_1^2$ | 4                        | 4                        | 4                       | 4                       | 4                       |

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A-T3-A

| $\mathcal{S}$ | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|

|                                                |                 |                |                |                |                 |
|------------------------------------------------|-----------------|----------------|----------------|----------------|-----------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$             | 4               | 4              | 4              | 4              | 4               |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$   | -4              | -4             | -4             | -4             | -4              |
| $ \lambda_5 ^2$                                | 12              | 12             | 12             | 12             | 12              |
| $g_2^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{9}{2}$  | $-\frac{9}{2}$ | $-\frac{9}{2}$ | $-\frac{9}{2}$ | $-\frac{9}{2}$  |
| $g_1^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{15}{2}$ | -3             | $-\frac{3}{2}$ | -3             | $-\frac{15}{2}$ |
| $\lambda_1 \lambda_{HS_1}^{(1)2}$              | 6               | 6              | 6              | 6              | 6               |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$          | 4               | 4              | 4              | 4              | 4               |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$     | 4               | 4              | 4              | 4              | 4               |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$      | 6               | 6              | 6              | 6              | 6               |
| $g_2^4$                                        | 0               | 0              | 0              | 0              | 0               |
| $g_1^4$                                        | 3               | $\frac{3}{4}$  | 0              | $\frac{3}{4}$  | 3               |
| $\lambda_{HS_1}^{(1)} T$                       | 2               | 2              | 2              | 2              | 2               |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$          | 0               | 0              | 0              | 0              | 0               |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$     | 0               | 0              | 0              | 0              | 0               |
| $\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$ | 0               | 0              | 0              | 0              | 0               |

Run block 2: T3-B-T3-B

| $\mathcal{S}$                                  | T3 - B ( $\alpha = -2$ ) | T3 - B ( $\alpha = -1$ ) | T3 - B ( $\alpha = 0$ ) | T3 - B ( $\alpha = 1$ ) | T3 - B ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$   | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                                | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^2 \lambda_{HS_1}^{(1)}$                   | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{15}{2}$          | -3                       | $-\frac{3}{2}$          | -3                      | $-\frac{15}{2}$         |
| $\lambda_1 \lambda_{HS_1}^{(1)2}$              | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                        | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                        | 3                        | $\frac{3}{4}$            | 0                       | $\frac{3}{4}$           | 3                       |
| $\lambda_{HS_1}^{(1)} T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$          | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 3: T3-C-T3-C

| $\mathcal{S}$                                | T3 - C ( $\alpha = -2$ ) | T3 - C ( $\alpha = -1$ ) | T3 - C ( $\alpha = 0$ ) | T3 - C ( $\alpha = 1$ ) | T3 - C ( $\alpha = 2$ ) |
|----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$           | 3                        | 3                        | 3                       | 3                       | 3                       |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$ | -3                       | -3                       | -3                      | -3                      | -3                      |
| $ \lambda_5 ^2$                              | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^2 \lambda_{HS_1}^{(1)}$                 | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_{HS_1}^{(1)}$                 | $-\frac{15}{2}$          | -3                       | $-\frac{3}{2}$          | -3                      | $-\frac{15}{2}$         |
| $\lambda_1 \lambda_{HS_1}^{(1)2}$            | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$        | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$   | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                      | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                      | 3                        | $\frac{3}{4}$            | 0                       | $\frac{3}{4}$           | 3                       |
| $\lambda_{HS_1}^{(1)} T$                     | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$        | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |

|                                                |   |   |   |   |   |
|------------------------------------------------|---|---|---|---|---|
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$     | 0 | 0 | 0 | 0 | 0 |
| $\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$ | 0 | 0 | 0 | 0 | 0 |

Run block 4: T3-D-T3-D

| $\mathcal{S}$                                  | T3 - D ( $\alpha = -2$ ) | T3 - D ( $\alpha = -1$ ) | T3 - D ( $\alpha = 0$ ) | T3 - D ( $\alpha = 1$ ) | T3 - D ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$   | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                                | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{33}{2}$          | $-\frac{33}{2}$          | $-\frac{33}{2}$         | $-\frac{33}{2}$         | $-\frac{33}{2}$         |
| $g_1^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{15}{2}$          | -3                       | $-\frac{3}{2}$          | -3                      | $-\frac{15}{2}$         |
| $\lambda_1 \lambda_{HS_1}^{(1)}$               | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^4$                                        | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_1^4$                                        | 3                        | $\frac{3}{4}$            | 0                       | $\frac{3}{4}$           | 3                       |
| $\lambda_{HS_1}^{(1)} T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$          | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$                                  | T3 - E ( $\alpha = -2$ ) | T3 - E ( $\alpha = -1$ ) | T3 - E ( $\alpha = 0$ ) | T3 - E ( $\alpha = 1$ ) | T3 - E ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$   | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                                | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{33}{2}$          | $-\frac{33}{2}$          | $-\frac{33}{2}$         | $-\frac{33}{2}$         | $-\frac{33}{2}$         |
| $g_1^2 \lambda_{HS_1}^{(1)}$                   | $-\frac{15}{2}$          | -3                       | $-\frac{3}{2}$          | -3                      | $-\frac{15}{2}$         |
| $\lambda_1 \lambda_{HS_1}^{(1)}$               | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^4$                                        | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_1^4$                                        | 3                        | $\frac{3}{4}$            | 0                       | $\frac{3}{4}$           | 3                       |
| $\lambda_{HS_1}^{(1)} T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$          | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$ | 4                        | 4                        | 4                       | 4                       | 4                       |

$\beta_{\lambda_{HS_2}^{(1)}}$

Run block 1: T3-A-T3-A

| $\mathcal{S}$                                | T3 - A ( $\alpha = -2$ ) | T3 - A ( $\alpha = -1$ ) | T3 - A ( $\alpha = 0$ ) | T3 - A ( $\alpha = 1$ ) | T3 - A ( $\alpha = 2$ ) |
|----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$           | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$ | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                              | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{HS_2}^{(1)}$                 | $-\frac{33}{2}$          | $-\frac{33}{2}$          | $-\frac{33}{2}$         | $-\frac{33}{2}$         | $-\frac{33}{2}$         |

|                                                |                |               |                 |                |                 |
|------------------------------------------------|----------------|---------------|-----------------|----------------|-----------------|
| $g_1^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{3}{2}$ | $-3$          | $-\frac{15}{2}$ | $-15$          | $-\frac{51}{2}$ |
| $\lambda_1 \lambda_{HS_2}^{(1)}$               | $6$            | $6$           | $6$             | $6$            | $6$             |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$          | $4$            | $4$           | $4$             | $4$            | $4$             |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$     | $8$            | $8$           | $8$             | $8$            | $8$             |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$      | $2$            | $2$           | $2$             | $2$            | $2$             |
| $g_2^4$                                        | $6$            | $6$           | $6$             | $6$            | $6$             |
| $g_1^4$                                        | $0$            | $\frac{3}{4}$ | $3$             | $\frac{27}{4}$ | $12$            |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$          | $2$            | $2$           | $2$             | $2$            | $2$             |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$     | $4$            | $4$           | $4$             | $4$            | $4$             |
| $\lambda_{HS_2}^{(1)} T$                       | $2$            | $2$           | $2$             | $2$            | $2$             |
| $\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$ | $0$            | $0$           | $0$             | $0$            | $0$             |

Run block 2: T3-B-T3-B

| $S$                                            | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$             | $2$                      | $2$                      | $2$                     | $2$                     | $2$                     |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$   | $-2$                     | $-2$                     | $-2$                    | $-2$                    | $-2$                    |
| $ \lambda_5 ^2$                                | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $g_2^2 \lambda_{HS_2}^{(1)}$                   | $-9$                     | $-9$                     | $-9$                    | $-9$                    | $-9$                    |
| $g_1^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{3}{2}$           | $-3$                     | $-\frac{15}{2}$         | $-15$                   | $-\frac{51}{2}$         |
| $\lambda_1 \lambda_{HS_2}^{(1)}$               | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$          | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$     | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$      | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $g_2^4$                                        | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                        | $0$                      | $\frac{3}{4}$            | $3$                     | $\frac{27}{4}$          | $12$                    |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$          | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$     | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |
| $\lambda_{HS_2}^{(1)} T$                       | $2$                      | $2$                      | $2$                     | $2$                     | $2$                     |
| $\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$ | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |

Run block 3: T3-C-T3-C

| $S$                                            | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$             | $3$                      | $3$                      | $3$                     | $3$                     | $3$                     |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$   | $-3$                     | $-3$                     | $-3$                    | $-3$                    | $-3$                    |
| $ \lambda_5 ^2$                                | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $g_2^2 \lambda_{HS_2}^{(1)}$                   | $-9$                     | $-9$                     | $-9$                    | $-9$                    | $-9$                    |
| $g_1^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{3}{2}$           | $-3$                     | $-\frac{15}{2}$         | $-15$                   | $-\frac{51}{2}$         |
| $\lambda_1 \lambda_{HS_2}^{(1)}$               | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$          | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$     | $6$                      | $6$                      | $6$                     | $6$                     | $6$                     |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$      | $4$                      | $4$                      | $4$                     | $4$                     | $4$                     |
| $g_2^4$                                        | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                        | $0$                      | $\frac{3}{4}$            | $3$                     | $\frac{27}{4}$          | $12$                    |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$          | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$     | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |
| $\lambda_{HS_2}^{(1)} T$                       | $2$                      | $2$                      | $2$                     | $2$                     | $2$                     |
| $\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$ | $0$                      | $0$                      | $0$                     | $0$                     | $0$                     |

Run block 4: T3-D-T3-D

| $\mathcal{S}$                                  | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$             | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$   | -4                       | -4                       | -4                      | -4                      | -4                      |
| $ \lambda_5 ^2$                                | 12                       | 12                       | 12                      | 12                      | 12                      |
| $g_2^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $g_1^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{3}{2}$           | -3                       | $-\frac{15}{2}$         | -15                     | $-\frac{51}{2}$         |
| $\lambda_1 \lambda_{HS_2}^{(1)}$               | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^4$                                        | 0                        | 0                        | 0                       | 0                       | 0                       |
| $g_1^4$                                        | 0                        | $\frac{3}{4}$            | 3                       | $\frac{27}{4}$          | 12                      |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$          | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$     | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{HS_2}^{(1)} T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$                                  | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|------------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$   | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                                | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{33}{2}$          | $-\frac{33}{2}$          | $-\frac{33}{2}$         | $-\frac{33}{2}$         | $-\frac{33}{2}$         |
| $g_1^2 \lambda_{HS_2}^{(1)}$                   | $-\frac{3}{2}$           | -3                       | $-\frac{15}{2}$         | -15                     | $-\frac{51}{2}$         |
| $\lambda_1 \lambda_{HS_2}^{(1)}$               | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_2^4$                                        | 6                        | 6                        | 6                       | 6                       | 6                       |
| $g_1^4$                                        | 0                        | $\frac{3}{4}$            | 3                       | $\frac{27}{4}$          | 12                      |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$          | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_2}^{(1)} T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$ | 4                        | 4                        | 4                       | 4                       | 4                       |

$\beta_{\lambda_{12}^{(1)}}$

Run block 1: T3-A-T3-A

| $\mathcal{S}$                                | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{12}^{(1)} T_\nu^{(1)}$             | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$ | -4                       | -4                       | -4                      | -4                      | -4                      |
| $ \lambda_5 ^2$                              | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{12}^{(1)}$                   | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 \lambda_{12}^{(1)}$                   | -6                       | -3                       | -6                      | -15                     | -30                     |
| $\left(\lambda_{12}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |

|                                               |   |               |   |                |    |
|-----------------------------------------------|---|---------------|---|----------------|----|
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$   | 4 | 4             | 4 | 4              | 4  |
| $g_2^4$                                       | 0 | 0             | 0 | 0              | 0  |
| $g_1^4$                                       | 0 | $\frac{3}{4}$ | 0 | $\frac{27}{4}$ | 48 |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$      | 4 | 4             | 4 | 4              | 4  |
| $\left(\lambda_{12}^{(A)}\right)$             | 0 | 0             | 0 | 0              | 0  |
| $\lambda_{12}^{(1)} T_\nu$                    | 0 | 0             | 0 | 0              | 0  |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$      | 0 | 0             | 0 | 0              | 0  |
| $\left(\lambda_{12}^{(\times)}\right)$        | 0 | 0             | 0 | 0              | 0  |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0 | 0             | 0 | 0              | 0  |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0 | 0             | 0 | 0              | 0  |

Run block 2: T3-B-T3-B

| $\mathcal{S}$                                 | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{12}^{(1)} T_\nu^{(1)}$              | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} T_\nu^{(2)}$              | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$  | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                               | 3                        | 3                        | 3                       | 3                       | 3                       |
| $g_2^2 \lambda_{12}^{(1)}$                    | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_{12}^{(1)}$                    | -6                       | -3                       | -6                      | -15                     | -30                     |
| $\left(\lambda_{12}^{(1)}\right)^2$           | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$      | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$   | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                       | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                       | 0                        | $\frac{3}{4}$            | 0                       | $\frac{27}{4}$          | 48                      |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)$             | $\frac{3}{4}$            | $\frac{3}{4}$            | $\frac{3}{4}$           | $\frac{3}{4}$           | $\frac{3}{4}$           |
| $\lambda_{12}^{(1)} T_\nu$                    | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(\times)}\right)$        | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 3: T3-C-T3-C

| $\mathcal{S}$                                | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{12}^{(1)} T_\nu^{(1)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} T_\nu^{(2)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$ | $-\frac{9}{2}$           | $-\frac{9}{2}$           | $-\frac{9}{2}$          | $-\frac{9}{2}$          | $-\frac{9}{2}$          |
| $ \lambda_5 ^2$                              | 3                        | 3                        | 3                       | 3                       | 3                       |
| $g_2^2 \lambda_{12}^{(1)}$                   | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_{12}^{(1)}$                   | -6                       | -3                       | -6                      | -15                     | -30                     |
| $\left(\lambda_{12}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$     | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$  | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                      | $\frac{9}{4}$            | $\frac{9}{4}$            | $\frac{9}{4}$           | $\frac{9}{4}$           | $\frac{9}{4}$           |
| $g_1^4$                                      | 0                        | $\frac{3}{4}$            | 0                       | $\frac{27}{4}$          | 48                      |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$     | 0                        | 0                        | 0                       | 0                       | 0                       |

|                                               |               |               |               |               |               |
|-----------------------------------------------|---------------|---------------|---------------|---------------|---------------|
| $\left(\lambda_{12}^{(A)}\right)^2$           | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$ |
| $\lambda_{12}^{(1)} T_\nu$                    | 3             | 3             | 3             | 3             | 3             |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$      | 0             | 0             | 0             | 0             | 0             |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0             | 0             | 0             | 0             | 0             |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0             | 0             | 0             | 0             | 0             |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0             | 0             | 0             | 0             | 0             |

Run block 4: T3-D-T3-D

| $\mathcal{S}$                                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{12}^{(1)} T_\nu^{(1)}$              | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{12}^{(1)} T_\nu^{(2)}$              | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$  | -4                       | -4                       | -4                      | -4                      | -4                      |
| $ \lambda_5 ^2$                               | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^2 \lambda_{12}^{(1)}$                    | -12                      | -12                      | -12                     | -12                     | -12                     |
| $g_1^2 \lambda_{12}^{(1)}$                    | -6                       | -3                       | -6                      | -15                     | -30                     |
| $\left(\lambda_{12}^{(1)}\right)^2$           | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$      | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$   | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                       | 0                        | 0                        | 0                       | 0                       | 0                       |
| $g_1^4$                                       | 0                        | $\frac{3}{4}$            | 0                       | $\frac{27}{4}$          | 48                      |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} T_\nu$                    | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$      | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$                                | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_{12}^{(1)} T_\nu^{(1)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} T_\nu^{(2)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$ | -2                       | -2                       | -2                      | -2                      | -2                      |
| $ \lambda_5 ^2$                              | $\frac{4}{3}$            | $\frac{4}{3}$            | $\frac{4}{3}$           | $\frac{4}{3}$           | $\frac{4}{3}$           |
| $g_2^2 \lambda_{12}^{(1)}$                   | -24                      | -24                      | -24                     | -24                     | -24                     |
| $g_1^2 \lambda_{12}^{(1)}$                   | -6                       | -3                       | -6                      | -15                     | -30                     |
| $\left(\lambda_{12}^{(1)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$     | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$  | 4                        | 4                        | 4                       | 4                       | 4                       |
| $g_2^4$                                      | 12                       | 12                       | 12                      | 12                      | 12                      |
| $g_1^4$                                      | 0                        | $\frac{3}{4}$            | 0                       | $\frac{27}{4}$          | 48                      |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\left(\lambda_{12}^{(A)}\right)^2$          | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_{12}^{(1)} T_\nu$                   | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$     | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$     | 4                        | 4                        | 4                       | 4                       | 4                       |

|                                               |   |   |   |   |   |
|-----------------------------------------------|---|---|---|---|---|
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 4 | 4 | 4 | 4 | 4 |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 4 | 4 | 4 | 4 | 4 |

$\beta_{\lambda_5}$

*Run block 1: T3-A-T3-A*

| $\mathcal{S}$                       | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_5 \lambda_1$               | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^2 \lambda_5$                   | $-\frac{21}{2}$          | $-\frac{21}{2}$          | $-\frac{21}{2}$         | $-\frac{21}{2}$         | $-\frac{21}{2}$         |
| $g_1^2 \lambda_5$                   | $-\frac{9}{2}$           | -3                       | $-\frac{9}{2}$          | -9                      | $-\frac{33}{2}$         |
| $\lambda_5 T_\nu^{(1)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 T_\nu^{(2)}$             | 1                        | 1                        | 1                       | 1                       | 1                       |
| $\lambda_5 \lambda_{HS_2}^{(A)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(A)}$    | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 \lambda_{12}^{(A)}$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T_\nu$                   | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

*Run block 2: T3-B-T3-B*

| $\mathcal{S}$                       | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_5 \lambda_1$               | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^2 \lambda_5$                   | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_5$                   | $-\frac{9}{2}$           | -3                       | $-\frac{9}{2}$          | -9                      | $-\frac{33}{2}$         |
| $\lambda_5 T_\nu^{(1)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T_\nu^{(2)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 \lambda_{HS_2}^{(A)}$    | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(A)}$    | -2                       | -2                       | -2                      | -2                      | -2                      |
| $\lambda_5 \lambda_{12}^{(A)}$      | $-\frac{1}{2}$           | $-\frac{1}{2}$           | $-\frac{1}{2}$          | $-\frac{1}{2}$          | $-\frac{1}{2}$          |
| $\lambda_5 T_\nu$                   | 1                        | 1                        | 1                       | 1                       | 1                       |
| $\lambda_5 \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

*Run block 3: T3-C-T3-C*

| $\mathcal{S}$                    | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|----------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_5 \lambda_1$            | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$ | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$ | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$   | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^2 \lambda_5$                | -9                       | -9                       | -9                      | -9                      | -9                      |
| $g_1^2 \lambda_5$                | $-\frac{9}{2}$           | -3                       | $-\frac{9}{2}$          | -9                      | $-\frac{33}{2}$         |
| $\lambda_5 T_\nu^{(1)}$          | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T_\nu^{(2)}$          | 0                        | 0                        | 0                       | 0                       | 0                       |

|                                     |                |                |                |                |                |
|-------------------------------------|----------------|----------------|----------------|----------------|----------------|
| $\lambda_5 \lambda_{HS_2}^{(A)}$    | 2              | 2              | 2              | 2              | 2              |
| $\lambda_5 T$                       | 2              | 2              | 2              | 2              | 2              |
| $\lambda_5 \lambda_{HS_1}^{(A)}$    | -2             | -2             | -2             | -2             | -2             |
| $\lambda_5 \lambda_{12}^{(A)}$      | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $-\frac{1}{2}$ |
| $\lambda_5 T_\nu$                   | $\frac{3}{2}$  | $\frac{3}{2}$  | $\frac{3}{2}$  | $\frac{3}{2}$  | $\frac{3}{2}$  |
| $\lambda_5 \lambda_{12}^{(\times)}$ | 0              | 0              | 0              | 0              | 0              |

Run block 4:  $T3-D-T3-D$

| $\mathcal{S}$                       | T3 - D ( $\alpha = -2$ ) | T3 - D ( $\alpha = -1$ ) | T3 - D ( $\alpha = 0$ ) | T3 - D ( $\alpha = 1$ ) | T3 - D ( $\alpha = 2$ ) |
|-------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_5 \lambda_1$               | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^2 \lambda_5$                   | $-\frac{21}{2}$          | $-\frac{21}{2}$          | $-\frac{21}{2}$         | $-\frac{21}{2}$         | $-\frac{21}{2}$         |
| $g_1^2 \lambda_5$                   | $-\frac{9}{2}$           | -3                       | $-\frac{9}{2}$          | -9                      | $-\frac{33}{2}$         |
| $\lambda_5 T_\nu^{(1)}$             | 1                        | 1                        | 1                       | 1                       | 1                       |
| $\lambda_5 T_\nu^{(2)}$             | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_2}^{(A)}$    | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(A)}$    | -4                       | -4                       | -4                      | -4                      | -4                      |
| $\lambda_5 \lambda_{12}^{(A)}$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T_\nu$                   | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 \lambda_{12}^{(\times)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5:  $T3-E-T3-E$

| $\mathcal{S}$                       | T3 - E ( $\alpha = -2$ ) | T3 - E ( $\alpha = -1$ ) | T3 - E ( $\alpha = 0$ ) | T3 - E ( $\alpha = 1$ ) | T3 - E ( $\alpha = 2$ ) |
|-------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $\lambda_5 \lambda_1$               | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$    | 4                        | 4                        | 4                       | 4                       | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $g_2^2 \lambda_5$                   | $-\frac{33}{2}$          | $-\frac{33}{2}$          | $-\frac{33}{2}$         | $-\frac{33}{2}$         | $-\frac{33}{2}$         |
| $g_1^2 \lambda_5$                   | $-\frac{9}{2}$           | -3                       | $-\frac{9}{2}$          | -9                      | $-\frac{33}{2}$         |
| $\lambda_5 T_\nu^{(1)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 T_\nu^{(2)}$             | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_5 \lambda_{HS_2}^{(A)}$    | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 T$                       | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(A)}$    | -2                       | -2                       | -2                      | -2                      | -2                      |
| $\lambda_5 \lambda_{12}^{(A)}$      | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_5 T_\nu$                   | 1                        | 1                        | 1                       | 1                       | 1                       |
| $\lambda_5 \lambda_{12}^{(\times)}$ | -2                       | -2                       | -2                      | -2                      | -2                      |

## Additional scalar-coupling beta functions

$$\beta_{\lambda_{HS_1}^{(A)}}$$

Run block 1:  $T3-A-T3-A$

| $\mathcal{S}$ | T3 - A ( $\alpha = -2$ ) | T3 - A ( $\alpha = -1$ ) | T3 - A ( $\alpha = 0$ ) | T3 - A ( $\alpha = 1$ ) | T3 - A ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 2: T3-B-T3-B

| $\mathcal{S}$ | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | -12                      | -6                       | 0                       | 6                       | 12                      |

Run block 3: T3-C-T3-C

| $\mathcal{S}$ | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | -12                      | -6                       | 0                       | 6                       | 12                      |

Run block 4: T3-D-T3-D

| $\mathcal{S}$ | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | -12                      | -6                       | 0                       | 6                       | 12                      |

Run block 5: T3-E-T3-E

| $\mathcal{S}$ | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | -12                      | -6                       | 0                       | 6                       | 12                      |

$$\beta_{\lambda_{HS_2}^{(A)}}$$

Run block 1: T3-A-T3-A

| $\mathcal{S}$ | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 6                        | 12                      | 18                      | 24                      |

Run block 2: T3-B-T3-B

| $\mathcal{S}$ | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 6                        | 12                      | 18                      | 24                      |

Run block 3: T3-C-T3-C

| $\mathcal{S}$ | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 6                        | 12                      | 18                      | 24                      |

Run block 4: T3-D-T3-D

| $\mathcal{S}$ | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$ | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 6                        | 12                      | 18                      | 24                      |

$$\beta_{\lambda_{12}^{(A)}}$$

Run block 1: T3-A-T3-A

| $\mathcal{S}$ | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 2: T3-B-T3-B

| $\mathcal{S}$ | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | -6                       | 0                       | 18                      | 48                      |

Run block 3: T3-C-T3-C

| $\mathcal{S}$ | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | -6                       | 0                       | 18                      | 48                      |

Run block 4: T3-D-T3-D

| $\mathcal{S}$ | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 5: T3-E-T3-E

| $\mathcal{S}$ | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|---------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_1^2 g_2^2$ | 0                        | -6                       | 0                       | 18                      | 48                      |

$$\beta_{\lambda_{12}^{(\times)}}$$

Run block 1: T3-A-T3-A

| $\mathcal{S}$                                 | T3 – A ( $\alpha = -2$ ) | T3 – A ( $\alpha = -1$ ) | T3 – A ( $\alpha = 0$ ) | T3 – A ( $\alpha = 1$ ) | T3 – A ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_2^4$                                       | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$  | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 2: T3-B-T3-B

| $\mathcal{S}$                                 | T3 – B ( $\alpha = -2$ ) | T3 – B ( $\alpha = -1$ ) | T3 – B ( $\alpha = 0$ ) | T3 – B ( $\alpha = 1$ ) | T3 – B ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_2^4$                                       | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$  | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

Run block 3: T3-C-T3-C

| $\mathcal{S}$                                 | T3 – C ( $\alpha = -2$ ) | T3 – C ( $\alpha = -1$ ) | T3 – C ( $\alpha = 0$ ) | T3 – C ( $\alpha = 1$ ) | T3 – C ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_2^4$                                       | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$  | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

*Run block 4: T3-D–T3-D*

| $\mathcal{S}$                                 | T3 – D ( $\alpha = -2$ ) | T3 – D ( $\alpha = -1$ ) | T3 – D ( $\alpha = 0$ ) | T3 – D ( $\alpha = 1$ ) | T3 – D ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_2^4$                                       | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$  | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | 0                        | 0                        | 0                       | 0                       | 0                       |

*Run block 5: T3-E–T3-E*

| $\mathcal{S}$                                 | T3 – E ( $\alpha = -2$ ) | T3 – E ( $\alpha = -1$ ) | T3 – E ( $\alpha = 0$ ) | T3 – E ( $\alpha = 1$ ) | T3 – E ( $\alpha = 2$ ) |
|-----------------------------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| $g_2^4$                                       | 6                        | 6                        | 6                       | 6                       | 6                       |
| $\left(\lambda_{12}^{(A)}\right)^2$           | 2                        | 2                        | 2                       | 2                       | 2                       |
| $\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$  | 8                        | 8                        | 8                       | 8                       | 8                       |
| $\left(\lambda_{12}^{(\times)}\right)^2$      | 10                       | 10                       | 10                      | 10                      | 10                      |
| $\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$ | –2                       | –2                       | –2                      | –2                      | –2                      |
| $\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$ | –2                       | –2                       | –2                      | –2                      | –2                      |

## First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether  $F$ ,  $S_1$ ,  $S_2$ , or a degenerate set is integrated out first.